

IOWA STATE UNIVERSITY

STRUCTURAL IDEALIZATION IN ANSYS

LAB ASSIGNMENT 05

MATTHEW MEHRTENS
LUCAS VASCONCELLOS

PROFESSOR

DR. BRANDON RUNNELS

College of Engineering
Aerospace Engineering
AERE 4210 Advanced Flight Structures

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INTRODUCTION

For part one, we were prompted to analyze an idealized stiffened panel structure consisting of a boom frame and a skin subjected to a distributed load applied at the free end (shown in [Figure 1.1](#)). The fixed end of the structure was constrained in all degrees of freedom. The material properties are $E = 70 \text{ GPa}$ and $\nu = 0.3$.

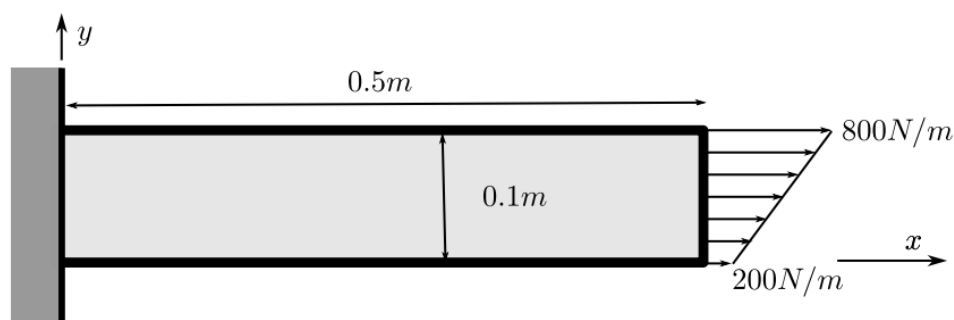


Figure 1.1: Stiffened panel dimensions and applied forces

For part two, we analyzed a tapered wing section composed of six stiffeners and connecting skins (shown in [Figure 1.2](#)). The vertical webs have a thickness of 1.0 mm , while the other skins have a thickness of 0.8 mm . The outer booms have a cross sectional area of 600 mm^2 , while the inner booms have a cross sectional area of 900 mm^2 . The material properties are $E = 70 \text{ GPa}$ and $\nu = 0.29$. The tapered end of the wing section was closed with a 10 mm thick sheet. The open end of the wing was constrained in all degrees of freedom.

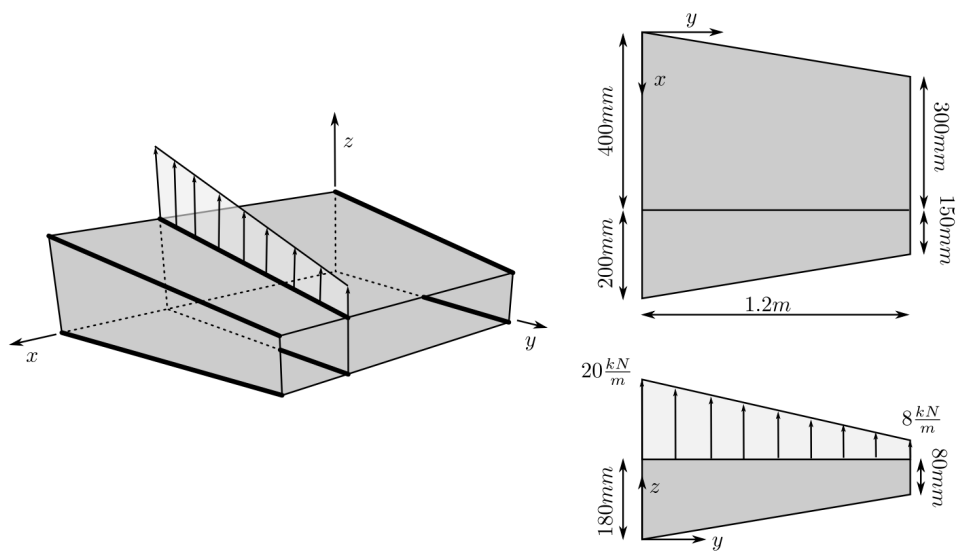


Figure 1.2: *Tapered wing section dimensions and applied forces*

RESULTS

2.1 Part 1

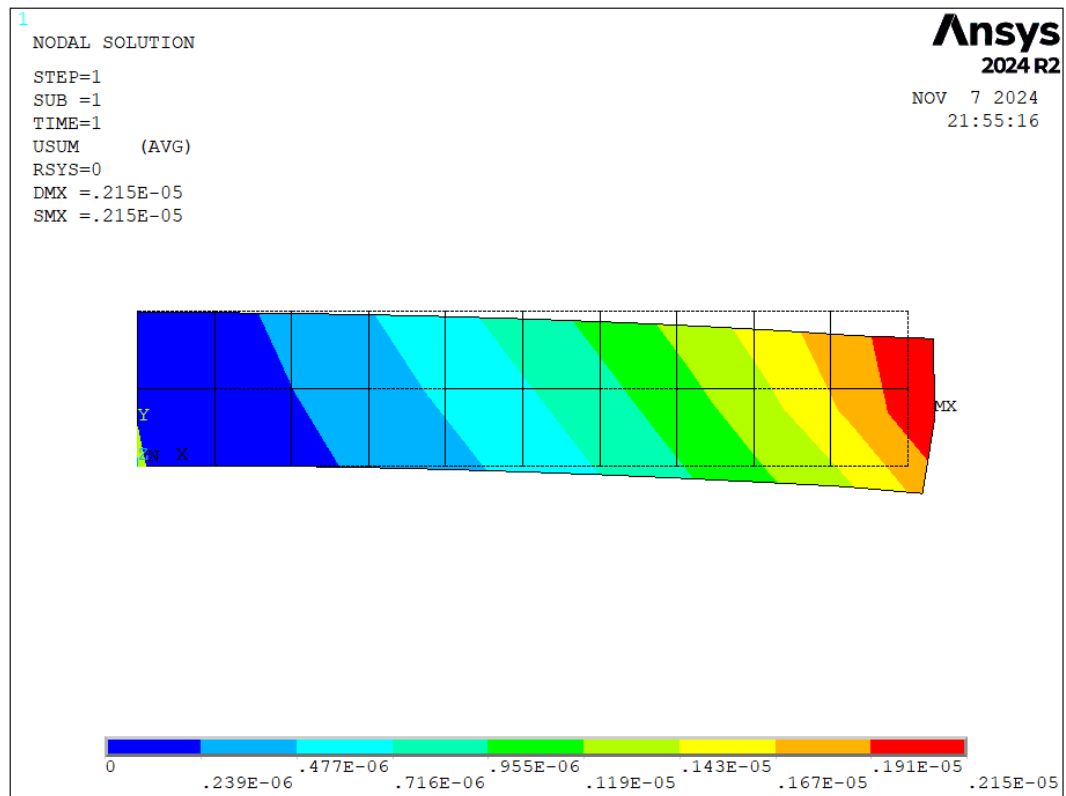


Figure 2.1: A contour plot of the total displacement.

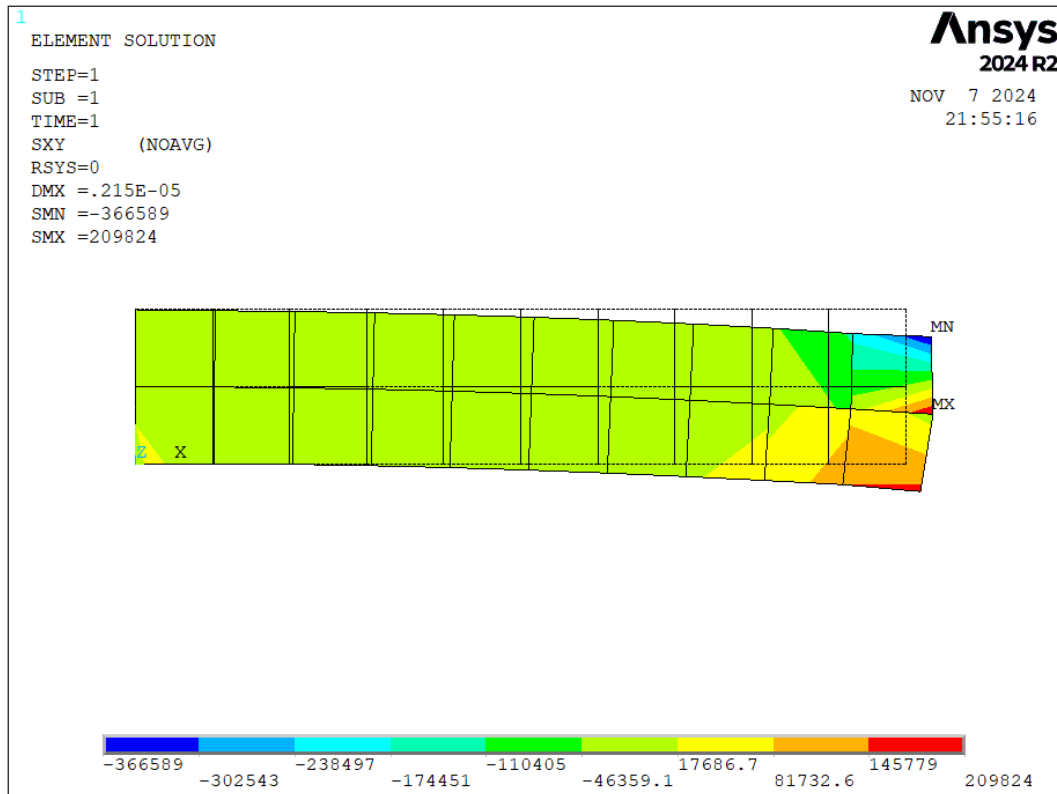


Figure 2.2: A contour plot of the xy-component of shear stress.

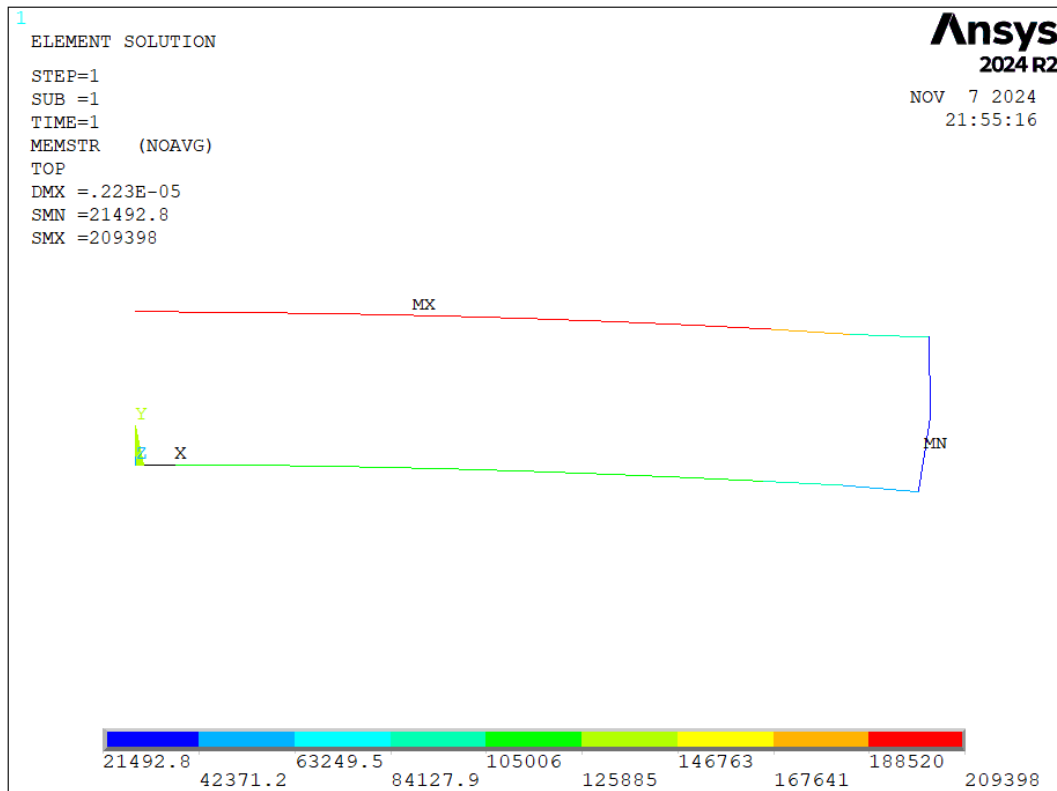


Figure 2.3: The member stress in each of the stiffeners.

2.2 Part 2

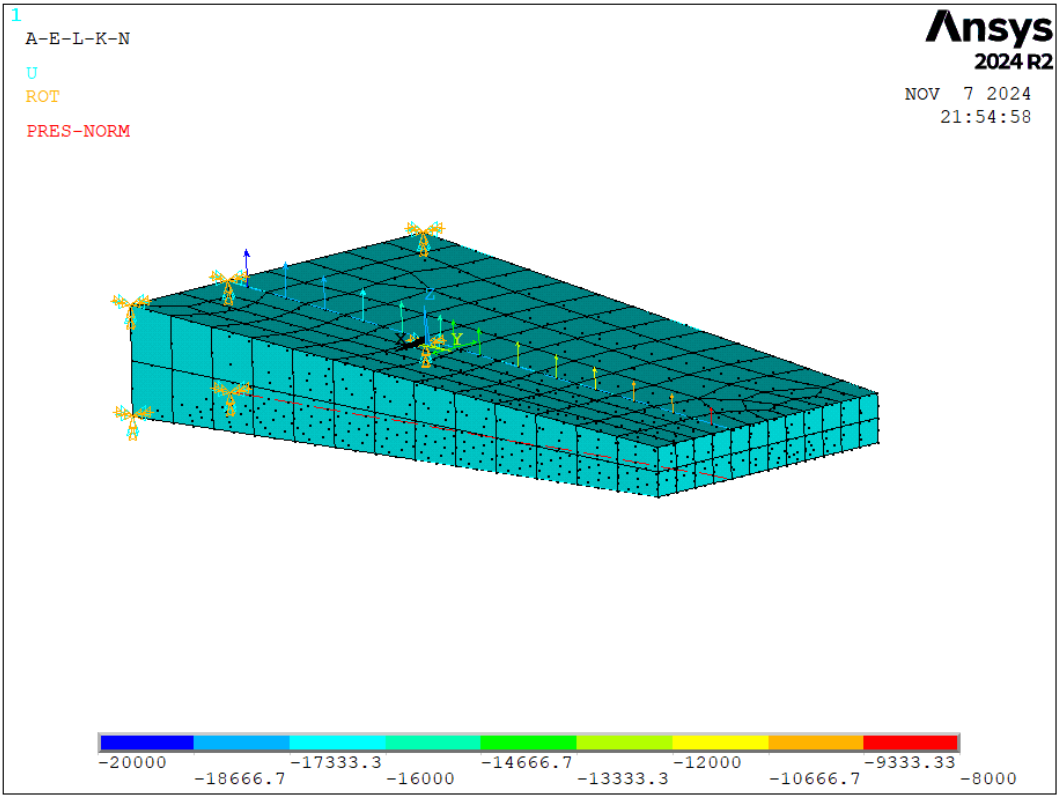


Figure 2.4: A schematic showing the geometry and loads.

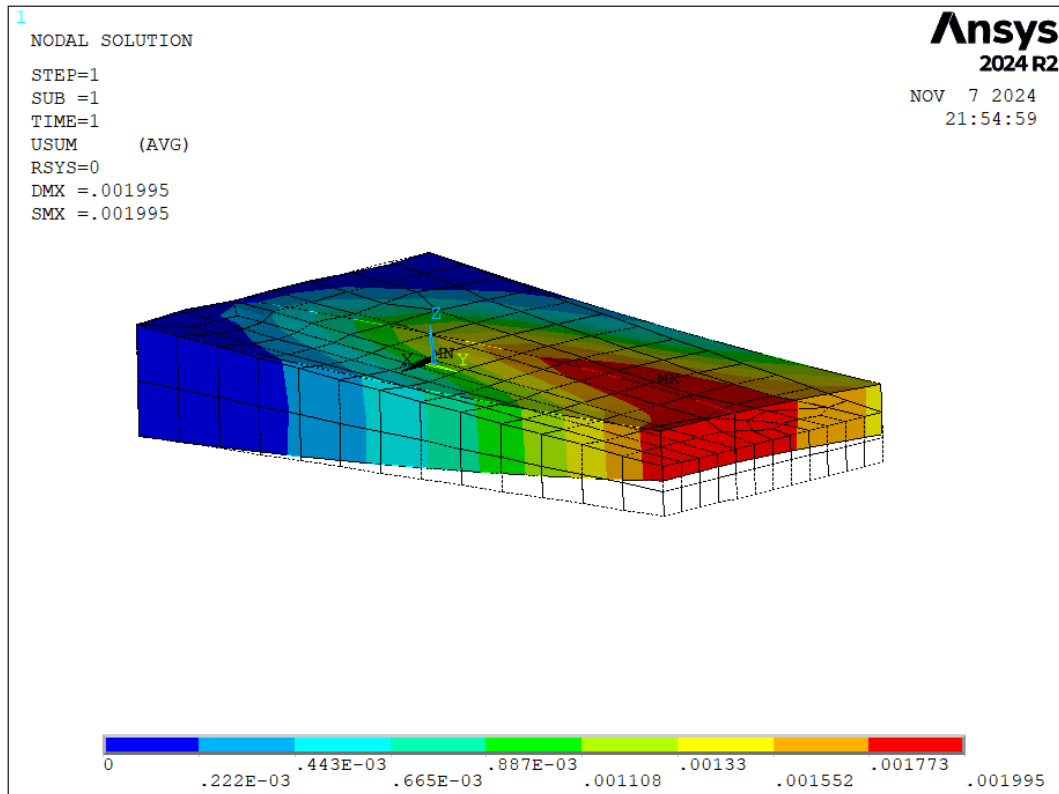


Figure 2.5: A contour plot of the total displacement.

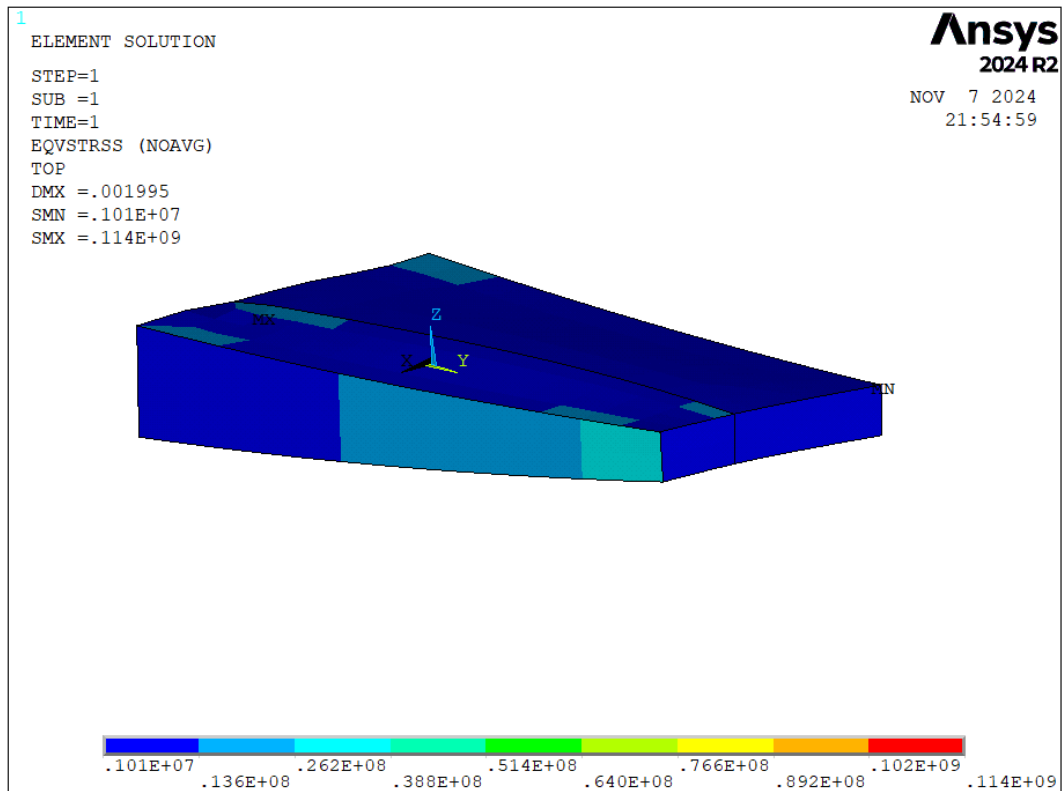


Figure 2.6: A visualization of the von Mises stress on the skins.

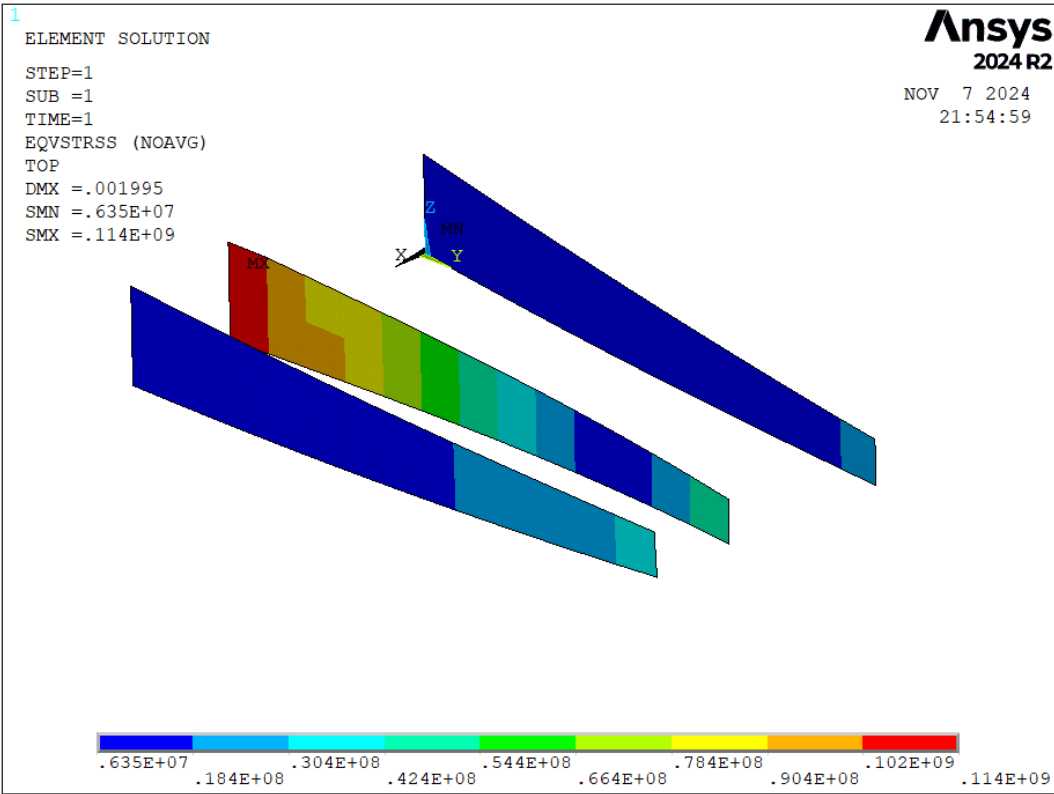


Figure 2.7: A visualization of the von Mises stress on the interior skins.

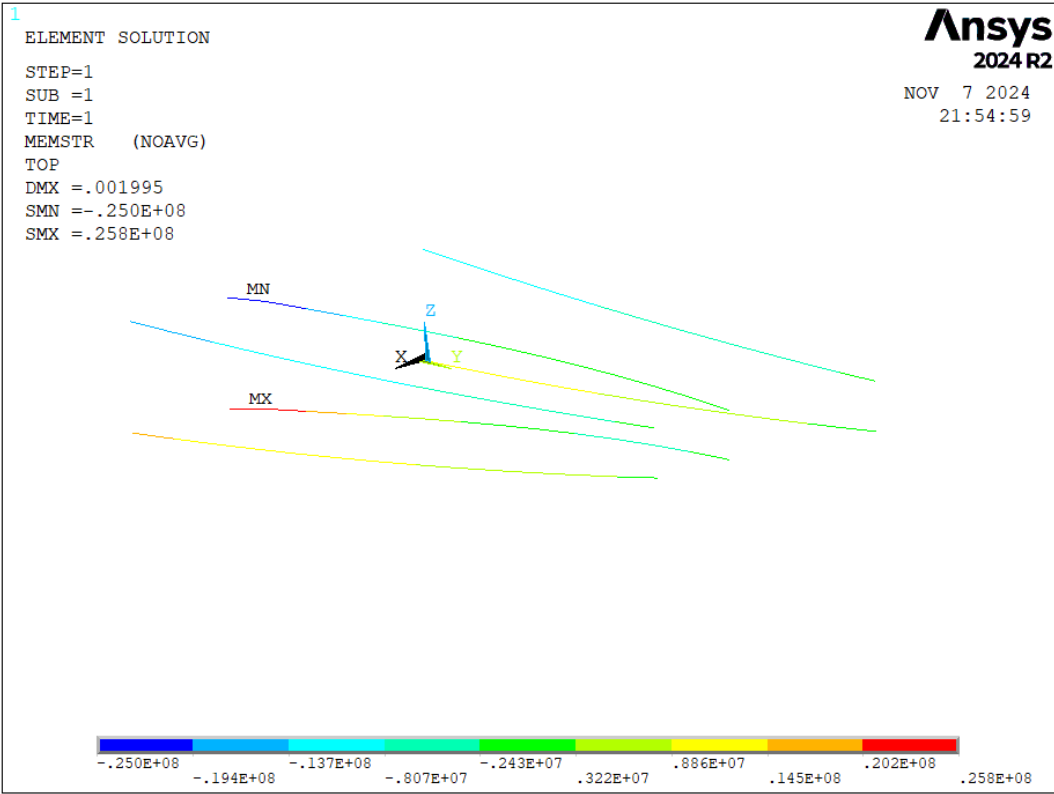


Figure 2.8: The member stress in each of the stiffeners.

CONCLUSION

The contour plots for part one were within what we expected for our results. The deformed shape and maximum deflection (see [Figure 2.1](#)) follows the expected output of a non-constant distributed load. The shear in the xy -plane of the skin concentrates a positive value at the bottom corner of the free end and a negative value at the top corner as shown in [Figure 2.2](#). As expected, in [Figure 2.3](#), we see on the free end the stiffeners have less stress. As the shear flows through the structure towards the fixed end, the member stress increases to the maximum.

In part two, the results match our intuition. The displacement (see [Figure 2.5](#)) is largest on the center stiffener where the distributed load is applied. The majority of the stress is carried by the center vertical skin, as seen in [Figure 2.7](#). The member stresses in [Figure 2.8](#) seem reasonable.

APPENDIX A

A.1 Part 1 APDL Script

```

/BATCH
/PREP7

! Define constants
*DIM,CWD,STRING,80
CWD(1) = 'U:\09-Fall2024\AERE4210\lab05\part1\'

! Reverse video
/RGB,INDEX,100,100,100, 0
/RGB,INDEX, 80, 80, 80,13
/RGB,INDEX, 60, 60, 60,14
/RGB,INDEX, 0, 0, 0,15
/REPLOT

! Define element type
ET,1,SHELL281
KEYOPT,1,8,1
ET,2,LINK180

! Define material properties
MP,EX,1,70e9
MP,EY,1,70e9
MP,EZ,1,70e9
MP,PRXY,1,0.3
MP,PRYZ,1,0.3
MP,PRXZ,1,0.3

! Define sections

```

```
SECTYPE,1,SHELL
SECDATA,1e-3,1,0,3
SECTYPE,2,LINK
SECDATA,1e-4

! Model the geometry
K,1,0,0
K,2,0.5,0
K,3,0.5,0.1
K,4,0,0.1
L,1,2 ! Line 1
L,2,3 ! Line 2
L,3,4 ! Line 3
L,4,1 ! Line 4
A,1,2,3,4
LSEL,S,LINE,,1,3,1,0
CM,STIFF,LINE

! Meshing
ASEL,S,AREA,,1
AATT,1,,1,,1
AMESH,ALL
CMSEL,S,STIFF
LATT,1,,2,,,2
LMESH,ALL

! Setting constraints
ALLSEL
DL,4,1,UX,0
DL,4,1,UY,0
DL,4,1,UZ,0
DL,4,1,ROTX,0
DL,4,1,ROTY,0
DL,4,1,ROTZ,0
SFL,2,PRES,-200,-800

! Solve
/SOLU
SOLVE

! Post process
/POST1
```

```
! Plot total displacement
PLNSOL,U,SUM,1,1.0
/IMAGE,SAVE,'%CWD(1)%total_displacement_contour_plot',png

! Plot xy stress
PLESOL,S,XY,1,1.0
/IMAGE,SAVE,'%CWD(1)%shear_xy',png

! Get stiffner stress
CMSEL,S,STIFF
ESLL,S

ETABLE,MEMSTR,LS,1
ETABLE,REFL
/OUTPUT,'%CWD(1)%memstr',txt
PRETAB,MEMSTR
/OUTPUT

PLETAB,MEMSTR,NOAV
/IMAGE,SAVE,'%CWD(1)%memstr',png
```

A.2 Part 2 APDL Script

```
/BATCH
/PREP7

! Define constants
*DIM,CWD,STRING,80
CWD(1) = 'U:\09-Fall2024\AERE4210\lab05\part2\'

! Reverse video
/RGB,INDEX,100,100,100, 0
/RGB,INDEX, 80, 80, 80,13
/RGB,INDEX, 60, 60, 60,14
/RGB,INDEX, 0, 0, 0,15
/REPLOT

! Define element type
ET,1,SHELL281
KEYOPT,1,8,1
ET,2,LINK180

! Define material properties
MP,EX,1,70e9
MP,EY,1,70e9
MP,EZ,1,70e9
MP,PRXY,1,0.29
MP,PRYZ,1,0.29
MP,PRXZ,1,0.29

! Define sections
SECTYPE,1,SHELL ! VERTSKIN
SECDATA,1.0e-3,1,0,3
SECTYPE,2,SHELL ! HORZSKIN
SECDATA,0.8e-3,1,0,3
SECTYPE,3,SHELL ! FRONTSKIN
SECDATA,10e-3,1,0,3
SECTYPE,4,LINK ! OUTERBOOM
SECDATA,6e-4
SECTYPE,5,LINK ! INNERBOOM
SECDATA,9e-4

! Model the geometry
```

K,1,0,0,0
K,2,100e-3,1.2,100e-3
K,3,100e-3,1.2,180e-3
K,4,0,0,180e-3

K,5,400e-3,0,0
K,6,400e-3,1.2,100e-3
K,7,400e-3,1.2,180e-3
K,8,400e-3,0,180e-3

K,9,600e-3,0,0
K,10,550e-3,1.2,100e-3
K,11,550e-3,1.2,180e-3
K,12,600e-3,0,180e-3

L,1,2 ! Line 1
L,4,3 ! Line 2
L,5,6 ! Line 3
L,8,7 ! Line 4
L,9,10 ! Line 5
L,12,11 ! Line 6

A,1,2,6,5 ! Area 1
A,5,6,10,9 ! Area 2
A,1,4,3,2 ! Area 3
A,2,3,7,6 ! Area 4
A,5,6,7,8 ! Area 5
A,6,7,11,10 ! Area 6
A,10,11,12,9 ! Area 7
A,3,4,8,7 ! Area 8
A,7,8,12,11 ! Area 9

LSEL,S,LINE,,1,2,1
LSEL,A,LINE,,5,6,1
CM,OUTERBOOM,LINE

LSEL,S,LINE,,3,4,1
CM,INNERBOOM,LINE

ASEL,S,AREA,,3,7,2
CM,VERTSKIN,AREA


```
ASEL,S,AREA,,1,2,1
ASEL,A,AREA,,8,9,1
CM,HORZSKIN,AREA
```

```
ASEL,S,AREA,,4,6,2
CM,FRONTSKIN,AREA
```

```
! Meshing
CMSEL,S,VERTSKIN
AATT,1,,1,,1
AMESH,ALL
```

```
CMSEL,S,HORZSKIN
AATT,1,,1,,2
AMESH,ALL
```

```
CMSEL,S,FRONTSKIN
AATT,1,,1,,3
AMESH,ALL
```

```
CMSEL,S,OUTERBOOM
LATT,1,,2,,,,4
LMESH,ALL
```

```
CMSEL,S,INNERBOOM
LATT,1,,2,,,,5
LMESH,ALL
```

```
! Setting constraints
KSEL,S,KP,,1,12,11
KSEL,A,KP,,4,5,1
KSEL,A,KP,,8,9,1
DK,ALL,UX,0
DK,ALL,UY,0
DK,ALL,UZ,0
DK,ALL,ROTX,0
DK,ALL,ROTY,0
DK,ALL,ROTZ,0
SFL,4,PRES,-20e3,-8e3
```

```
! Transfer to FE
ASEL,S,AREA,,5
```

SBCTRAN

```
! Rotate and capture geometry
/VIEW,1,1,0,0
/ANGLE,1,-90,XM,1
/ANGLE,1,-45,ZM,1
/VIEW,1,0.697522674336,0.690215639509,0.192521400795
/ANGLE,1,-100.792326989
/REPLOT
/VIEW,1,0.661382760771,0.720717138681,0.207700866068
/ANGLE,1,-102.217371015
/REPLOT
/VIEW,1,0.623449237286,0.734418906039,0.268216179565
/ANGLE,1,-106.458594546
/PSF,PRES,NORM,2,0,ON
/PBC,ALL,,1
/REPLOT
GPLOT
/IMAGE,SAVE,'%CWD(1)%geometry',png
```

```
! Solve
/SOLU
SOLVE
```

```
! Post process
/POST1
```

```
! Get maximum displacement
PLNSOL,U,SUM,1,1.0
/IMAGE,SAVE,'%CWD(1)%displacement',png
```

```
! Get von Mises stress on skins
ESEL,S,TYPE,,1
ETABLE,EQVSTRSS,S,EQV
PLETAB,EQVSTRSS,NOAV
/IMAGE,SAVE,'%CWD(1)%von_Mises-all',png
```

```
ESEL,S,SEC,,1
ETABLE,REFL
/ANGLE,1,15,XS,1
PLETAB,EQVSTRSS,NOAV
/IMAGE,SAVE,'%CWD(1)%von_Mises-interior',png
```

```
/ANGLE,1,-15,XS,1

! Get stresses on the stiffeners
ESEL,S,TYPE,,2
ETABLE,MEMSTR,LS,1
PLETAB,MEMSTR,NOAV
/IMAGE,SAVE,'%CWD(1)%memstr',png
```